

SCS 1307
Probability & Statistics

by
Dr Dilanthi Gamage Dona
Department of Statistics
University of Colombo

Normal Distribution

Normal distribution is the most important continuous distribution in statistics. Many measured quantities in the natural sciences follow a normal distribution.

eg. heights, masses, ages, examination results.

A continuous random variable X having p.d.f

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}.$$

is said to have a normal distribution with mean μ and variance σ^2 .

μ and σ^2 are the parameters of the distribution.

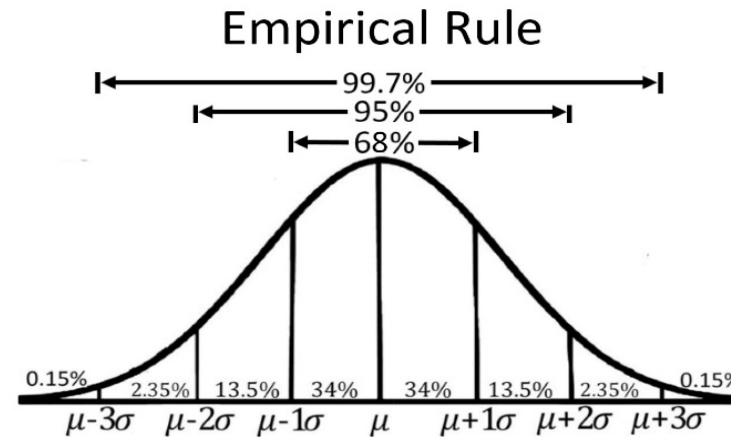
The distribution is bell shaped and symmetrical about $x = \mu$.

If $X \sim N(\mu, \sigma^2)$ then

$$E(X) = \mu \text{ and}$$

$$\text{Var}(X) = \sigma^2$$

Empirical Rule



Area under the total curve is 1

Approximately 68% of the distribution lies within $\pm 1 \sigma$ of the mean.

Approximately 95% of the distribution lies within $\pm 2 \sigma$ of the mean.

Approximately 99.8% of the distribution lies within $\pm 3 \sigma$ of the mean.

Consider the random variable Z where $Z = \frac{(X - \mu)}{\sigma}$

Now if the random variable X has a normal distribution with mean μ and variance σ^2 , then the random variable Z has a standard normal distribution with mean 0 and variance 1.

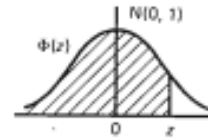
The p.d.f. of the standard normal variable X is denoted by $\phi(X)$ where

$$\phi(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}x^2}.$$

This integral is difficult to evaluate, so we refer to tables.

$$\Phi(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^x e^{-t^2/2} dt$$

THE DISTRIBUTION FUNCTION $\Phi(z)$ OF
THE NORMAL DISTRIBUTION $N(0, 1)$



z	0	1	2	3	4	5	6	7	8	9	1	2	3	4	5	6	7	8	9
											ADD								
0.0	.5000	.5040	.5080	.5120	.5160	.5199	.5239	.5279	.5319	.5359	4	8	12	16	20	24	28	32	36
0.1	.5398	.5438	.5478	.5517	.5557	.5596	.5636	.5675	.5714	.5753	4	8	12	16	20	24	28	32	36
0.2	.5793	.5832	.5871	.5910	.5948	.5987	.6026	.6064	.6103	.6141	4	8	12	15	19	23	27	31	35
0.3	.6179	.6217	.6255	.6293	.6331	.6368	.6406	.6443	.6480	.6517	4	7	11	15	19	22	26	30	34
0.4	.6554	.6591	.6628	.6664	.6700	.6736	.6772	.6808	.6844	.6879	4	7	11	14	18	22	25	29	32
0.5	.6915	.6950	.6985	.7019	.7054	.7088	.7123	.7157	.7190	.7224	3	7	10	14	17	20	24	27	31
0.6	.7257	.7291	.7324	.7357	.7389	.7422	.7454	.7486	.7517	.7549	3	7	10	13	16	19	23	26	29
0.7	.7580	.7611	.7642	.7673	.7704	.7734	.7764	.7794	.7823	.7852	3	6	9	12	15	18	21	24	27
0.8	.7881	.7910	.7939	.7967	.7995	.8023	.8051	.8078	.8106	.8133	3	5	8	11	14	16	19	22	25
0.9	.8159	.8186	.8212	.8238	.8264	.8289	.8315	.8340	.8365	.8389	3	5	8	10	13	15	18	20	23
1.0	.8413	.8438	.8461	.8485	.8508	.8531	.8554	.8577	.8599	.8621	2	5	7	9	12	14	16	19	21
1.1	.8643	.8665	.8686	.8708	.8729	.8749	.8770	.8790	.8810	.8830	2	4	6	8	10	12	14	16	18
1.2	.8849	.8869	.8888	.8907	.8925	.8944	.8962	.8980	.8997	.9015	2	4	6	7	9	11	13	15	17
1.3	.9032	.9049	.9066	.9082	.9099	.9115	.9131	.9147	.9162	.9177	2	3	5	6	8	10	11	13	14
1.4	.9192	.9207	.9222	.9236	.9251	.9265	.9279	.9292	.9306	.9319	1	3	4	6	7	8	10	11	13
1.5	.9332	.9345	.9357	.9370	.9382	.9394	.9406	.9418	.9429	.9441	1	2	4	5	6	7	8	10	11
1.6	.9452	.9463	.9474	.9484	.9495	.9505	.9515	.9525	.9535	.9545	1	2	3	4	5	6	7	8	9
1.7	.9554	.9564	.9573	.9582	.9591	.9599	.9608	.9616	.9625	.9633	1	2	3	4	4	5	6	7	8
1.8	.9641	.9649	.9656	.9664	.9671	.9678	.9685	.9693	.9699	.9706	1	1	2	3	4	4	5	6	6
1.9	.9713	.9719	.9726	.9732	.9738	.9744	.9750	.9756	.9761	.9767	1	1	2	2	3	4	4	5	5
2.0	.9772	.9778	.9783	.9788	.9793	.9798	.9803	.9808	.9812	.9817	0	1	1	2	2	3	3	4	4
2.1	.9821	.9826	.9830	.9834	.9838	.9842	.9846	.9850	.9854	.9857	0	1	1	2	2	2	3	3	4
2.2	.9861	.9864	.9868	.9871	.9875	.9878	.9881	.9884	.9887	.9890	0	1	1	1	2	2	2	3	3
2.3	.9893	.9896	.9898		.9901	.9903	.9906	.9908			0	1	1	1	1	2	2	2	2
								.9911	.9913	.9915	3	5	8	10	13	15	18	20	23
2.4	.9918	.9920	.9922	.9924	.9926						2	5	7	9	12	14	16	18	21
					.9928	.9930	.9932	.9934	.9936		2	4	6	8	11	13	15	17	19
2.5	.9937	.9939	.9941	.9943	.9944	.9946	.9947	.9949	.9950	.9952	2	3	5	6	8	9	11	12	14
2.6	.9953	.9954	.9956	.9957	.9958	.9959	.9960	.9961	.9962	.9963	1	2	3	5	6	7	8	9	10
2.7	.9965	.9966	.9967	.9968	.9969	.9970	.9971	.9972	.9973	.9974	1	2	3	4	5	6	7	8	9
2.8	.9974	.9975	.9976	.9977	.9978	.9979	.9980	.9981	.9982	.9983	1	1	2	3	4	4	5	6	6
2.9	.9983	.9984	.9985	.9986	.9987	.9988	.9989	.9990	.9991	.9992	0	1	1	2	2	3	3	4	4
3.0	.9993	.9994	.9995	.9996	.9997	.9998	.9999	.9999	.9999	.9999	0	1	1	2	2	2	3	3	4
3.1	.9999	.9999	.9999	.9999	.9999	.9999	.9999	.9999	.9999	.9999	3	6	9	13	16	19	22	25	28
			.9999	.9999	.9999	.9999	.9999	.9999	.9999	.9999	3	6	8	11	14	17	20	22	25
3.2	.9999	.9999	.9999	.9999	.9999	.9999	.9999	.9999	.9999	.9999	2	5	7	10	12	15	17	20	22
								.9999	.9999	.9999	2	4	7	9	11	13	15	18	20
3.3	.9999	.9999	.9999	.9999	.9999	.9999	.9999	.9999	.9999	.9999	2	4	6	8	9	11	13	15	17
								.9999	.9999	.9999	2	3	5	6	8	10	11	13	14
3.4	.9999	.9999	.9999	.9999	.9999	.9999	.9999	.9999	.9999	.9999	1	3	4	5	7	8	9	10	12
								.9999	.9999	.9999	1	2	3	4	5	6	7	8	9
3.5	.9999	.9999	.9999	.9999	.9999	.9999	.9999	.9999	.9999	.9999	1	1	2	3	4	4	5	6	7
3.6	.9999	.9999	.9999	.9999	.9999	.9999	.9999	.9999	.9999	.9999	0	1	1	2	2	3	3	4	5
3.7	.9999	.9999	.9999	.9999	.9999	.9999	.9999	.9999	.9999	.9999									
3.8	.9999	.9999	.9999	.9999	.9999	.9999	.9999	.9999	.9999	.9999									
3.9	.9999	.9999	.9999	.9999	.9999	.9999	.9999	.9999	.9999	.9999									

For negative values of z use $\Phi(z) = 1 - \Phi(-z)$